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THE HISTORY OF

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SQUEAK

Back to the future: the PARC Smalltalk lineage reborn as free,
self-hosting, media-rich software in 1996

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THE BAND GETS BACK TOGETHER

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By the mid-1990s the original Learning Research Group ideas were locked inside commercial Smalltalks licensed by the seat. Alan Kay's team at Apple -- Dan Ingalls, Ted Kaehler, John Maloney, Scott Wallace -- wanted a Smalltalk they could give to children, researchers, and themselves: free, portable, and malleable all the way down. Their solution, described in the OOPSLA '97 paper "Back to the Future: The Story of Squeak, A Practical Smalltalk Written in Itself" [1], was an act of software archaeology and bootstrapping at once.

They started from the Apple Smalltalk-80 image, then wrote the new virtual machine in a subset of Smalltalk itself (Slang) which a Smalltalk program translates to C. The result: the VM that runs the image is written and debugged inside the image it runs. Squeak was released in 1996, months before the team moved to Disney Imagineering; because Apple's license made it free with sources, it spread immediately to every platform with a C compiler.

MORPHIC, ETOYS, AND CHILDREN AGAIN

Squeak absorbed Morhic, the direct-manipulation UI framework from Self (Maloney and Randall Smith's earlier work), making every on-screen object grabbable, inspectable, and scriptable while the system runs. On top of Morhic the team built Etoys, a tile-scripting environment in the Dynabook tradition (see that chapter): children script cars, animations, and simulations by dragging tiles, with the full Smalltalk beneath the tiles when they outgrow them.

Etoys shipped on the One Laptop per Child XO-1, putting a direct descendant of the PARC vision in the hands of millions of children -- the closest the literal Dynabook program ever came to its intended audience. Squeak also carried Kay's media ambitions: real-time sound synthesis, MIDI, movies, 3D (Croquet and later Open Cobalt built shared 3D worlds on it), and network media, all inside one image.

COMMUNITY, FORKS, AND CONTINUITY

After Disney, stewardship passed to a worldwide community (squeak.org [2]), with Viewpoints Research Institute -- Kay's nonprofit -- using Squeak as the substrate for experiments like the STEPS project (reinventing personal computing in 20,000 lines). The relicensing to MIT/Apache cleared the way for inclusion in Linux distributions. Croquet, Seaside (the continuation-based web framework), and Scratch -- MIT's blocks language for children, whose first generations were implemented in Squeak -- are all its descendants or tenants.

Squeak remains actively developed (releases through the 6.x line), still delivered as a bit-identical image that opens with

the same live objects on every operating system. Its most consequential fork, Pharo (2008), gets its own chapter: where Squeak stayed loyal to media and education, Pharo aimed the same substrate at professional software engineering. Together they kept the original PARC system -- not a reimplementaion, the actual continuously-evolved image -- alive into the 2020s: the oldest living codebase of the personal-computing revolution.

SOURCES & CITATIONS

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- [1] Dan Ingalls, Ted Kaehler, John Maloney, Scott Wallace, Alan Kay: Back to the Future: The Story of Squeak, A Practical Smalltalk Written in Itself, OOPSLA 1997
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- [2] Squeak official site: downloads, documentation, community
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- [3] Squeak/Smalltalk history resources and weekly news, including the BYTE archive pointer
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- [4] Etoys for the One Laptop per Child project
<http://www.squeakland.org/>
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